

ANNUAL REPORT



National Academy of
Agricultural Research Management
Rajendranagar, Hyderabad - 500 030, India

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1996-97



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FOREWORD

National Academy of Agricultural Research Management was established in 1976 to organize foundation courses for the entry level Agricultural Research Service scientists; and training programmes, workshops, seminars and conferences for the middle and senior level scientists representing Indian National Agricultural Research System (NARS) of national and international agencies on all aspects of agricultural research, education management and extension technology. NAARM also acts as a repository of high quality resource material in the field of agricultural research and education management based on case / research / field studies. The Academy is also assisting in extending technical guidance on agricultural research and education management to the State Agricultural Universities and establishing educational technology centres which support teaching and research community in enhancing their capabilities in imparting knowledge and skills appropriately. This annual report presents the events and activities of 1996-97.

During this year, NAARM organized a number of training programmes - foundation, specialized and advanced courses; and workshops for various levels of scientists of ICAR, SAUs, DOA, CSB, etc. For all the organized programmes, the NAARM adopted a unique system of matrix model.

I acknowledge with appreciation the efforts put in for compiling and editing the manuscript by Drs K. Venkateswarlu, Jagannadham Challa, T. Balaguru and Shri R.V.V.S. Prakasa Rao.

August 18, 1997.

J. C. Katyal
(J.C. Katyal)
Director



INTRODUCTION

Over the last four decades, Indian agriculture has recorded remarkable increase in agricultural production and has acquired considerable resilience and stability, largely through generation of better technology in land and water management, input use, control of pests and pathogens, and adoption of scientific production techniques. This success has underlined the need for a better management of our research education and transfer of technology systems to optimize production in agricultural sector.

Realizing the need for an institutional set-up to impart training in agricultural research project management and for adopting principles of educational technology for better education, the National Commission on Agriculture (NCA) recommended the establishment of an all India institute for administration and management in agriculture. The National Academy of Agricultural Research Management (NAARM), earlier known as Central Staff College for Agriculture, was thus set up in 1976 as one of the constituent institutions of the Indian Council of Agricultural Research (ICAR) under the aegis of the Department of Agricultural Research and Education (DARE), Ministry of Agriculture, Govt of India.

The Academy, since its inception, has been assisting the deemed-to-be Universities, Research Institutes of the ICAR, and the State Agricultural Universities (SAUs) by providing training input for management of agricultural research, education, training, and extension.

The Academy is located in the sylvan environs of Rajendranagar, in the premises of Acharya N.G. Ranga Agricultural University (ANGRAU), on the south-western periphery of Hyderabad city. The well-planned campus, spreading over 50 hectares, has several modern buildings including academic block, faculty-cum-laboratory building, administrative block, halls of residence, scientists' home, agro-technology demonstration blocks, and several faculty and staff residences.

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Mandate

- To organize and conduct training programmes in agricultural research management for the scientists at various levels.
- To build up high quality resource material in agricultural research management based on actual field experience.
- To undertake systematic review and study of management problems of agricultural research Institutes, programmes and systems.
- To plan, organize and conduct workshops and seminars in research management and educational technology.
- To organize, liaise and co-ordinate programmes of international co-operation in the field of agricultural research management.
- To assist the State Agricultural Universities (SAUs) in developing Regional Centers of Management to cater to the needs of various Universities.

Objectives

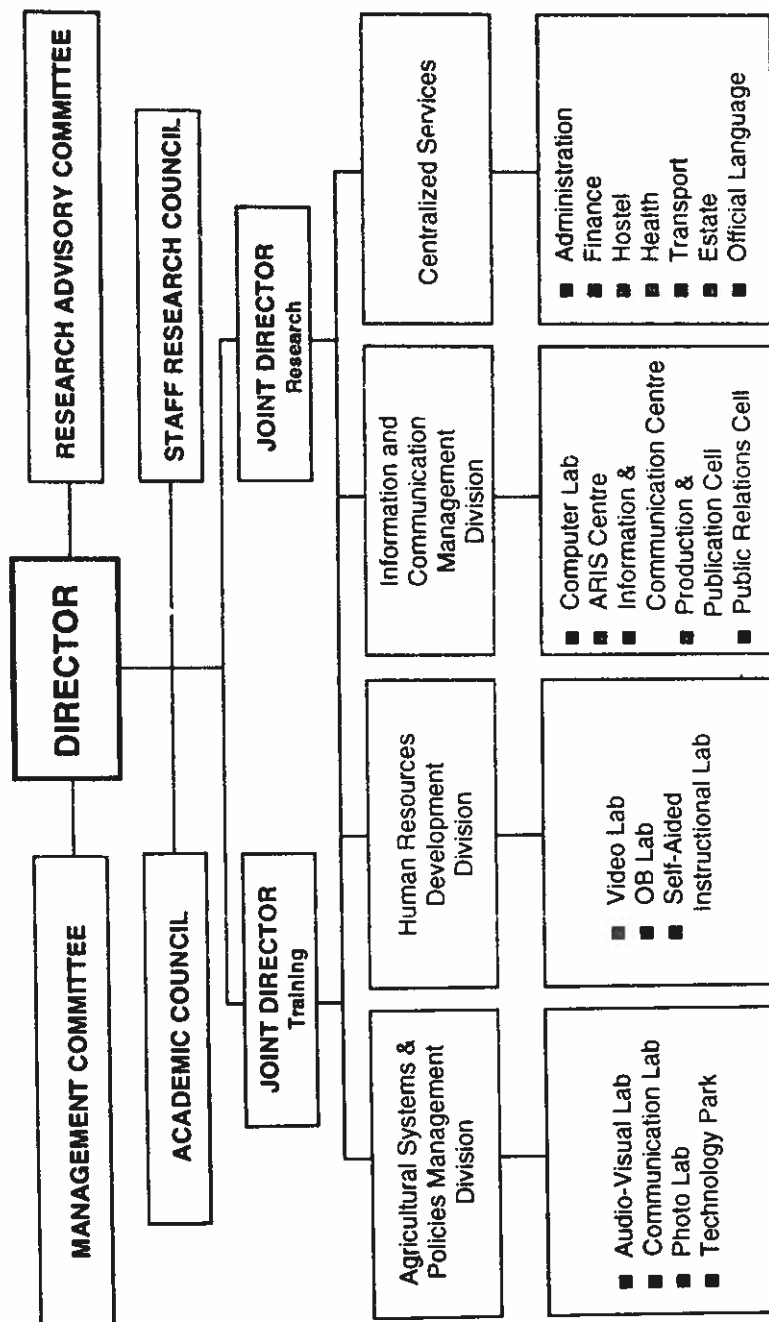
- To organize foundation courses for the new entrants of scientists to the Agricultural Research Service of ICAR.
- To design and organize training programmes, workshops and seminars; and develop high quality material in agricultural research and education management for the senior and middle level scientists of NARS.
- To play a major role in different processes of human resource development of NARS in the context of agricultural research and education.
- To undertake policy level studies on prioritization of research and resources allocation, and to enhance research productivity of the constituents of NARS and their programmes.
- To evolve suitable management techniques for effective planning, scheduling, organizing, monitoring and evaluation of agricultural research and development projects in NARS.

- * To study various programmes and patterns of technology transfer, develop effective systems and strategies for effective communication of agricultural technologies.
- * To undertake development programmes for evolving more effective and efficient approaches, methods and models in higher agricultural education, and to provide support to SAUs and ICAR Institutes in agricultural education.
- * To promote and execute academic programmes of higher learning in the specialized areas of agricultural research and education management
- * To develop expertise in information technology to meet the application software needs of NARS.
- * To undertake systematic review and develop appropriate systems for efficient and effective administrative and financial management in NARS.
- * To function as a repository of ideas and information in the form of databases in the field of agricultural research and education management, and to act as a clearing house for information dissemination through national and international networks in these areas.
- * To offer consultancy services in the field of agricultural research and education management, agricultural information and communication technology.

Organisational structure

The organisational structure of NAARM is given in the figure, as indicated in the Perspective Plan. The Director is the Chief Executive of NAARM and is guided by the Management Committee (MC). The MC reviews and promotes programmes of the Academy; and approves new areas of research, education, training programmes, workshops and seminars. In the area of research, the Research Advisory Committee (RAC) and Staff Research Council (SRC) provide broad guidelines and assist in developing and implementing specific programmes and projects. In the execution of various programmes the Director is assisted by two Joint Directors of Research and Training who, in turn, assisted by three Division Heads and centralized services. In organizing various programmes - training, workshops and seminars, NAARM follows matrix model.

Fig. ORGANIZATIONAL STRUCTURE OF NAARM



GLIMPSES

2

1996

April

- A National Training-cum-Workshop on Agricultural Research Financial Information System (ARFIS) was organized, by the ICAR, at the Academy from April 15 to 23, 1996, in three spells. Around 300 financial and administrative personnel of the ICAR system attended the training programme.
- An advanced Course on Agricultural Research Project Management was organized from April 16 to 26, 1996. Twenty five scientists from various ICAR Institutes, SAUs, and other organizations participated in the course.
- A Course on Educational Video Production was organized from April 23 to May 4, 1996. Twenty scientists from ICAR Institutes participated in the course.

June

- A Summer Institute on Recent Advances in Agricultural Research Management was organized from June 3 to 22, 1996. Twenty seven scientists, in the rank of professors and associate professors, principal and senior scientists from SAUs and ICAR Institutes participated in the programme.
- A Course on Management Information System (MIS) for Agricultural Research was organized from June 11 to 22, 1996. Thirteen senior scientists and teachers working in the ICAR Institutes, SAUs, and other research institutes participated in the course.
- The third contact session of the Outreach Programme in Educational Technology (OPET) was organized from June 18 to 28, 1996. Eighteen faculty members from five SAUs participated in the programme.

July

- A Course on Computer Applications in Agricultural Research was organized from July 15 to 22, 1996. Twenty six scientists from ICAR Institutes and SAUs attended the course.

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■ Estate
■ Official Language

■ Publication Cell
■ Public Relations Cell

■ Institutional Law

■ Technology Park

August

- A Faculty Development Programme in Educational Technology for the faculty members of Rajasthan Agricultural University, Bikaner, was organized from August 5 to 20, 1996, at the Directorate of Extension, College of Agriculture, Udaipur.
- First Programme on Payroll/GPF Packages was organized from August 6 to 8, 1996. Twenty six financial and administrative personnel of the ICAR system participated in the programme.

September

- A National Workshop on PR-oriented Marketing Communication Strategies of Agricultural Research was organized from September 2 to 7, 1996. Twenty four PR professionals belonging to ICAR institutes, SAUs, and Banking sector, including business executives and entrepreneurs from private sector, participated in the workshop.
- A Course on Videography in Agriculture was organized from September 3 to 13, 1996. Seventeen technical personnel engaged in videography and photography in ICAR Institutes and SAUs, and also the scientific staff looking after these sections attended the course.
- Second programme on Payroll / GPF Packages was organized from September 9 to 13, 1996. Twenty eight financial and administrative personnel of ICAR institutes participated in the programme.
- A Training Programme on UNESCO's CDS/ISIS (Ver 3.07) and Associated Packages was organized from September 20 to October 1, 1996. Twenty seven participants who include librarians, documentalists, information specialists, analysts, and technical personnel, handling information services in the ICAR Institutes, SAUs, and other related organizations, participated in this course.
- A Course on Manpower Planning in Agriculture was organized from September 19 to 27, 1996. Four scientists from ICAR Institutes and SAUs participated in the course.
- A Course on Application of SWOT Analysis in NARS was organized from September 23 to 26, 1996. Twenty eight scientists from ICAR Institutes and SAUs attended the course.

- An International Management Development Programme on Management of Agricultural Research was organized from September 30 to October 11, 1996. Twenty five Directors of various Agricultural Research Institutes from Nigeria participated in the programme.

October

- Faculty Development Programme in Educational Technology was organized for the faculty members of Gujarat Agricultural University from October 2 to 24, 1996 at Anand.

November

- An advanced Course on Management of Human Resources for Research was organized from November 18 to 30, 1996. Eleven participants who include scientists, teachers, and administrators of the ICAR Institutes, and other R & D and educational institutes participated in the programme.
- A Course on Statistical Softwares for Data Analysis was organized from November 19 to 29, 1996. Eighteen scientists and officers from various ICAR Institutes and SAUs, who are involved in data collection and analysis, participated in the programme.

December

- A Workshop on Official Language Management (Hindi) was organized from December 17 to 19, 1996. Twenty five officers and staff working in the Hindi Sections of the ICAR Institutes participated in the workshop.

1997

January

- A programme on Payroll / GPF Packages was organized from January 3 to 9, 1997. Sixteen financial and administrative personnel of the ICAR system participated in the programme.
- An Orientation and Refresher Course for TANUVAS faculty was organized from

January 16 to 31, 1997. Fifteen faculty members of Tamilnadu University of Veterinary and Animal Sciences, Chennai participated in the course.

February

- A programme on Payroll /GPF Packages was organized from February 13 to 19, 1997. Twenty two financial and administrative personnel of the ICAR system participated in the programme.

March

- A programme on Educational Management and Administration for TANUVAS faculty was organized from March 5 to 14, 1997. Fifteen faculty members of TANUVAS, Chennai participated in the programme.
- A Course on Video Production for Transfer of Technology in Agriculture was organized from March 10 to 22, 1997. Five scientists and Technical officers of ICAR Institutes participated in the course.

PROGRAMMES AND ACTIVITIES

3

The Academy has been planning, organizing and conducting various training programmes on different aspects of management relating to agricultural research, education, extension education, and training for the junior, middle and senior level scientists of ICAR Institutes and SAUs, which form the two major areas of the National Agricultural Research System (NARS). It also conducts training programmes for executives, senior research managers, and administrative and accounts officers. Research studies carried out by the faculty members on the functioning of the research and educational institutions and programmes in the country provide realistic data for supporting the Academy's programmes and for identifying the management problems including policy issues.

3.1 Training

Training is an important input for Human Resources Development in all categories of the human resource-mix engaged in teaching, research and extension functions. Agricultural research and education management is one such important sub-sector. The Academy imparts training in the areas of research management, educational technology, and training management for various levels of scientists of the ICAR, SAUs, and other public and private sector organizations including the NGOs. The training programmes have been grouped into two main sub-heads viz., Foundation Course for Agricultural Research Service (FOCARS) for probationers; Specialized and Advanced Training programmes including Workshops, Seminars, and Conferences for senior level scientists and policy makers.

3.1.1 Foundation Course

Foundation course is designed for the scientists recruited to Agricultural Research Service of the ICAR. It is the most important programme of the Academy. The course aims at providing exposure to the trainees on the concepts and principles of project management with special emphasis on project formulation and implementation. It also includes instructions in related areas on human resources development and management, transfer of technology systems and policies, and information management. The specific objectives of the FOCARS are as follows:

Objectives

- To appraise scientist trainees about agricultural scenario, and the agricultural research, education and development systems in the country.
- To sensitize the scientist trainees about the global agricultural scenarios and its implications to Indian agriculture.
- To stimulate critical thinking and multidisciplinary approach to agricultural research.
- To expose scientist trainees to modern concepts of agricultural research project formulation, management methods and processes, and promote competency.
- To develop knowledge and skills in scientific communication.
- To understand modern concepts and techniques of information management and use of information technology to enhance scientific input in agricultural research.
- To enable scientist trainees to understand their behaviour and style of work, and its implications in research organization, with special emphasis on personality development, inter-personal relationship, team work, dynamics of human motivation, and organizational behaviour.
- To expose scientist trainees to the socioeconomic milieu of rural society in general, and farming community in particular, and to develop empathy for appropriate technology generation and farmers and appreciation of the need to innovate technology diffusion and adoption.

The course is structured into three phases. The first phase of training is at the Academy, which concentrates mainly on the basics of Agricultural Research Management, Organizational Behaviour, Transfer of Technology, Information Management, Administration and Accounts Procedures. The second phase, Field Experience Training (FET), is held at the regional stations of SAUs and ICAR Institutes. The FET is designed to impart training in project identification and formulation. This phase helps the trainees develop an appreciation for rural life and environment. In the third phase of training at the Academy, the trainees are given an exposure to those areas which would strengthen their skills in scientific writing and presentation, and improvement of communication skills.

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Dr S.K. Nanda, Co-ordinator of the 59th FOCARS addressing the probationers. Also seen (L to R) Dr S.N. Saha, Jt Director; J.C. Katyal, Director and Dr P. Kumar another Co-ordinator of the course.



Drawing social map by ARS probationers during their field experience training.

The overall programme also includes educational visits to various research institutes in and around Hyderabad to familiarize them with different organizations and the research activities. During the period under report, 101 ARS scientists, freshly recruited into the service, completed their training. Currently 35 ARS scientists are undergoing training at the Academy. The details are given in table 1.

Table 1. Foundation Course for Agricultural Research Service (FOCARS) Scientists

S.No.	Course Number	Period	Co-ordinators	No. of Participants
FOCARS				
1.	56	05.07.96 to 02.12.96	Dr D. Rama Rao Dr K. Vidyasagar Rao	39
2.	57	28.08.96 to 25.01.97	Dr VE. Sabarathnam Dr (Ms) R. Kalpana Sastry	36
3.	58	07.01.97 to 06.05.97	Dr K. Venkateswarlu Dr B.S. Chandel	26
4.	59	04.03.97 to 01.07.97	Dr P. Kumar Dr S.K. Nanda	45

3.1.2 Specialized and Advanced Courses

Research management training is a continuous process of learning and updating skills. The Academy offers various specialized programmes to all categories of scientists. These specialized programmes essentially aimed at developing an appreciation for acquiring appropriate management skills by providing excellent opportunities among the scientists at various levels from ICAR and SAUs. The trainees of senior courses include Principal Scientists, Professors, Heads of Divisions/Departments, Directors of Research, Project Coordinators, and Directors of Institutes. During the year under report, the faculty had to their credit a record of over 21 programmes, including two international training programmes, (table 2), which are described below:

Agricultural Research Financial Information System (ARFIS)

The programme was organized to familiarize the ICAR accounts personnel with the features and operation of a computerized accounting software package developed for computerization of monthly accounts of all ICAR Institutes, with the main objective of standardization of accounting procedures and for uniform classification. With this package, the entire financial system of the ICAR would get uniformly standardized and the working would get further simplified.

Educational Video Production

The course exposed the participants to the concepts of educational video programmes and their creative aspects, develops the skills for programme production, and introduces the participants to the new and emerging areas of video application. Through extensive hands-on experience, the participants were assisted in developing good institutional video tapes in their respective areas of specialization.

Computer Applications in Agricultural Research

The programme was organized to create an awareness among the scientists of ICAR Institutes and SAUs on the applications of computers, and agricultural data storage, updating, processing and retrieval, word processing packages such as Word Star and Microsoft Word and other packages like D-Base III+ and Harvard Graphics. The course provided excellent opportunities for the participants to practise these packages using the modern computer facilities available. This course is in greater demand from the scientists of ICAR and SAUs.

Payroll / GPF Packages

The programme was organized to familiarize the ICAR accounts personnel with the features and operation of the computerized accounting software package on Payroll/GPF to have uniform structure all over the ICAR, not only from the funding source point of view but also each sub-headwise which helps in inter-facing with ARFIS package. Further, it helps in storage and retrieval of financial information, drawl of arrears, etc.

Videography in Agriculture

The programme was organized with a view to promoting communication technologies in the research institutions of ICAR and SAUs, exposing the participants to the technical aspects of videography, and introducing them to the emerging areas of videographic applications. Some of the topics covered include video recording system and format, principles of cinematography and photography, working of video cameras and VCR, studio and field recording, lighting in studios etc.

CDS/ISIS and Associated Packages

The programme was organized to discuss the concept of databases, DBMS and designing of bibliographic databases; to provide hands-on experience on the use of CDS/ISIS package; and to expose the library information management personnel to the application programmes related to CDS/ISIS. The application programmes covered during the training programme along with CDS/ISIS (ver 3.07) are: CAS & SDI services; generation of library card catalogues and accession register; thesaurus generation and use in CDS/ISIS; downloading data from CD-ROM to CDS/ISIS; data export from dBase to CDS/ISIS and *vice versa*. The other related packages covered include SANJAY, TRISHNA, HEURISCO, Windows version of CDS/ISIS, etc.

SWOT Analysis of NARS

The SWOT analysis allows the organizations to develop and build up strengths, overcome weaknesses, prevent threats, and exploit opportunities based on the situation at a given time. Keeping in view the importance of the application of SWOT analysis for agricultural research organizations, the course was organized to expose the participants to the concept and use of SWOT for enhancing their research effectiveness. Case studies were used for discussion on various topics.

Table 2 Specialized and Advanced Courses			
Sl. No.	Training Programme	Co-ordinators	No. of participants
A. Specialized Courses			
1.	National Training on Agril. Res. Financial Information System(ARFIS) in three spells	Mr B.Pandu Reddy Mr U. Muralidhar	April 15-23 300
2.	Educational Video Production	Mr K.R.Prabhakar	April 23 - May 4 20
3.	Computer Applications in Agricultural Research	Mr U.Muralidhar Dr K.V.S. Rao	July 15-27 22
4.	Payroll / GPF Packages (1st programme)	Mr U.Muralidhar Mr B.Pandu Reddy Mr M.Suresh Kumar	Aug. 6-8 26
5.	Videography in Agriculture	Mr K.R.Prabhakar Mr Ch.Janardhan Rao	Sept. 3-13 17
6.	Payroll / GPF Packages (2nd Programme)	Mr U.Muralidhar Mr B.Pandu Reddy Mr M.Suresh Kumar	Sept. 9-13 28
7.	CDS/ISIS and Associated Packages	Mr Y.V.Sivaprasad Mr U.Muralidhar	Sept. 20-Oct. 1 27
8.	Application of SWOT Analysis in NARS	Dr K.Venkateswarlu Dr J.C. Kalla	Sept. 23-26 28
9.	Statistical Softwares for Data Analysis	Dr K.V.S. Rao Mr U.Muralidhar	Nov. 19-29 18
10.	Payroll / GPF Packages (3rd Programme)	Mr B.Pandu Reddy Mr M.Suresh Kumar Mr M.J.Pradeep Kumar	Dec. 5-11 22
11.	Payroll / GPF Packages (4th Programme)	Mr U.Muralidhar Mr B.Pandu Reddy Mr M.Suresh Kumar	Jan. 3-9 16

No. of participants	
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12. Orientation and Refresher Course for TANUVAS Faculty	Dr K.M. Reddy Dr A. Gopalam Dr M.M. Anwer	Jan. 16-31	15
13. Payroll / GPF Packages (5th Programme)	Mr U. Muralidhar Mr B. Pandu Reddy Mr M. Suresh Kumar	Feb. 13-19	22
14. Educational Management and Administration for TANUVAS Faculty	Dr K.M. Reddy Mr M.Suresh Kumar Dr A. Gopalam	Mar. 5-14	15
15. Video Production for Transfer of Technology in Agriculture	Mr K.R. Prabhakar Mr Ch. Janardhan Rao	Mar. 10-22	5
B. Advanced Courses			
1. Agricultural Research Project Management	Dr S.N. Saha Dr K.Venkateswarlu	Apr. 16-26	25
2. Management Information System for Agricultural Research(MIS)	Dr T. Balaguru Dr P. Manikandan Dr J.C. Kalla	June 11-22	13
3. Manpower Planning in Agriculture	Dr VE.Sabarathnam	Sept. 19-27	4
4. Management of Human Resources for Res. Managers	Dr M.M. Anwer Dr R.Kalpana Sastry	Nov. 18-30	11
C. International Training Programmes			
1. International Management Development Programme on Management of Agril. Res. for Nigerian Directors (NAARM & ARMTI, Nigeria)	Dr J.C. Kalla, C.E.O. Dr T. Balaguru Dr M.M. Anwer Dr P. Manikandan	Sept. 30 - Oct. 11, 1996 &	25
	Dr S.N. Saha, C.E.O. Dr T. Balaguru Dr D. Rama Rao Mr M.Suresh Kumar	Feb. 3-14, 1997	27

Statistical Softwares for Data Analysis

The course was organized to provide an exposure to the basics of computer, to familiarize the scientists to work on statistical software packages for data analysis, and to provide adequate training on presentation of data and interpretation of results from the analysis. The course covered various areas like basic concepts of DOS operating system, hardware and software; statistical software packages of SPSS, MSTATC, MICROSTAT, Lotus 1-2-3; Harvard graphics; and other related topics like GIS, remote sensing, GIST, etc.

Video Production for Transfer of Technology in Agriculture

The course was organized to develop skills in the production of educational video programmes, as video has become a very important tool in the transfer of technology in agriculture. The video tapes are becoming increasingly important in the present day agricultural extension systems owing to their ability to attract and sustain the attention of the viewers. The course exposed them, including hands-on experience, on various aspects of instructional video production.

Agricultural Research Project Management

The course exposed the research managers to the application of management concepts to agricultural research projects for ensuring their relevance and efficient implementation. Various aspects of research project cycle; viz., planning, priority setting, formulation, monitoring and evaluation; and other related topics such as management of differences and brainstorming were covered through information sharing, group discussion, case analysis, and exercises.

Management Information System for Agricultural Research

The course was conducted with a view to acquainting the scientists with the concept of Management Information System (MIS) for agricultural research and assisting them to develop a computer-based MIS towards better research management in agriculture. As a part of the overall MIS, the course also exposed the participants to a project-based budgeting system (PBS). Through a project work, the participants prepared the MIS for a small research institute and generated various reports for managerial decision making.

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Dr S.N. Saha, Acting Director, NAARM addressing the participants of Paryroll/GPF Package. To his right (L to R) Mr M. Suresh Kumar, Mr U. Muralidhar and to his left B. Pandu Reddy, Programme Co-ordinator.



Dr T. Balaguru, Programme Co-ordinator, addressing the delegates. Also seen (L to R) Mr M. Suresh Kumar, Programme Co-ordinator; Dr Olu Makinde, Director, ARMTI, Nigeria; Dr S.L. Mehta, Dy. Director (Edn), ICAR; Dr S.N. Saha, Acting Director, NAARM; Dr O.A. Odegbare, Consultant, NARP, Nigeria; and Dr D. Rama Rao, Programme Co-ordinator.

Manpower Planning in Agriculture

The course aimed at imparting skills to organize, plan and utilize efficiently the manpower engaged in research, teaching, extension, and administration to achieve the objectives of the agricultural research organization. The course covered details on manpower planning, time management, manpower utilization, etc.

Management of Human Resources for Research

The course was organized to expose the research managers to the concepts of human behaviour in organizations (organizational behaviour) and to sensitize them to the issues in management of human resources. It also aimed at improving the awareness of senior scientists in this vital area for helping them to effectively manage people at work, thereby increasing the productivity of research.

International Programmes

International Management Development Programme on Management of Agricultural Research

Two programmes were organized for the Directors of Agricultural Research Institutes from Nigeria, focusing on aspects like research programme planning, priority setting, monitoring and evaluation; human resources, financial and administrative management; computer application and MIS; implementation of NARP and its impact in India and Nigeria, etc.

The entire programme was divided into two phases. In the first phase, the delegates interacted with specially identified experts, in addition to the resource persons from NAARM, who made presentations on various topics related to the management of agricultural research. In the second phase, the delegates were on a study visit, which mainly focused on identifying various managerial issues in running the research institutes in India. Some of the institutes they visited were Rubber Research Institute (RRI), Kottayam; IIHR, Bangalore; CPCRI, Kasaragod; CTCRI, Trivandrum; IARI, New Delhi; IVRI, Izatnagar; NDRI, Karnal; CIFA, Bhubaneswar and CMFRI, Cochin.

3.2 Research

The Academy is endowed with a mandate to conduct research in the areas of agricultural research and education management. It carries out research in various areas such as agricultural policy, research project management, transfer of technology, human resources development, agricultural education, and information technology. A brief description of various research projects undertaken is presented below :

1. **A profile of Scientific Human Resources in ICAR System in India** (*S. Shanmugam*)

The project was undertaken to study the socio-economic attributes of cadre strength of scientific personnel in ICAR Institutes, and to plan the optimum human resources profile in ICAR system for 2001 A.D. The data collection is in progress.

2. **Developing Suitable Management Information System (MIS) for NARS** (*T. Balaguru, P. Manikandan, Jagdeesh C. Kalla and K. Vidyasagar Rao*)

The project was taken up to develop a computer-based methodology for the creation and management of MIS, and to improve the effectiveness of research in agricultural research institutions.

The study resulted in the publication of basic MIS methodology in the form of a Book for adoption by the research institutions. Besides, three case studies pertaining to one ICAR Institute and two research stations of an Agricultural University were prepared. These were used as resource materials in the senior level course on Management Information System for Agricultural Research.

3. **Evolving a Suitable Methodological Framework for Priority Setting in Agriculture** (*T. Balaguru*)

The project was initiated with the objective of developing suitable methodology for setting research priorities in agriculture. Action was initiated to collect data from a few selected research institutes.

4. Development and Analysis of Database on Crops, Forestry, Livestock and Fishery Resources in NARS
(K. Vidyasagar Rao & P. Kumar)

The data collected earlier on livestock population and productivity have been analysed and interpreted. The period of study was divided into three parts of five years each (1980-84, 1985-89 and 1990-94)

A comparison between goat and buffalo population data revealed that increase in goats population was higher than buffaloes during two periods i.e. 1985 to 94 in India as well as in the world. Similarly, there was a gradual increase in milk production during the same period in the global situation as well as in India. The production of poultry meat was also increased one and half a times during the last period i.e., 1990-94. Egg production was also increased in India at a greater pace during the last period over the second period. The research findings are under print.

5. Estimation and Evaluation of Students Performance of Agricultural Graduates of Andhra Pradesh Agricultural University, Hyderabad
(K. Vidyasagar Rao, Jagdeesh C. Kalla and K.V. Kumar)

The study revealed that the parental occupation had no impact on the student performance in Intermediate. This suggests a thorough review of the present reservation policy based on parental occupation to make reservation criteria more meaningful. The results revealed that the level of standard at Intermediate had positive and linear impact on the overall, grade point average (OGPA) in B.Sc (Ag). In such a case, the question has to be answered whether the present selection process for undergraduation in Agriculture, through common entrance test (EAMCET), is necessary as an alternative to test the student caliber. The results on performance during graduation indicated that the variability in semester course among batches is highly random for both male and female students. This phenomenon could be mainly attributed to the creeping of high subjectivity in the evaluation system. This really causes an alarm and needs special attention of the administrators and policy makers.

6. Technological Forecasting of Groundnut Scenario in India
(D. Rama Rao and Kiresur)

The major objectives of the project are to forecast oilseeds scenario in the country and beyond using technological forecasting methods; to analyse various factors /

components / variables contributing to the oilseeds situation using econometric and systems methods; and to provide policy guidelines for strategic planning on oilseeds. Final report of the research project was submitted.

7. Professional (Graduates) Women in Agriculture (PWA)
(D. Rama Rao, Ms M. Suguna Sri)

The Project was undertaken to study the status and growth of agricultural women graduates; career opportunities, professional performances, and identification of issues pertaining to women professionals and the influencing factors.

Three different questionnaires were designed and sent to equal number of male and female respondents. Out of the 6248 women identified, it was proposed to reach 30% i.e., about 2000 women and 2000 men. The data collection is in progress.

8. Management and Impact of Agricultural Research Stations / Institutes in India
(K. Venkateswarlu)

The project aims at studying the processes and details of technology developed in agricultural research stations and also the productivity of various research/ experimental stations. The constraints faced in the management of research stations would also be found out. Case studies have been initiated and data are being collected for this purpose. The work is under progress.

a) Criteria for Selection and Formulation of R & D Projects in Agriculture and Allied Areas
(K. Venkateswarlu and S.K. Nanda)

The project aims to study the criteria for selection of R & D projects in agriculture and in allied areas, and also to study the formulation process of various R & D projects. The data collection is in progress.

b) Creativity and Problem Solving Nature of Agricultural Scientists
(K. Venkateswarlu)

The main objective of this study is to understand the creativity and problem solving

nature of scientists in developing new theories / inventions / discoveries in the field of agricultural sciences. The data have been collected from the scientists and senior scientists through a standard questionnaire. The work is under progress.

9. Management and Impact of All India Co-ordinated Research Projects
(K. Venkateswarlu and S. Shanmugam)

The major objectives of the project are to identify the lacunae in the structure and organization of the All India Co-ordinated Research Projects (AICRPs) and also to study the impact of various AICRPs. A questionnaire has already been designed and mailed to the project co-ordinators to collect the data on these aspects. The data collection is in progress.

10. Impact of National Agricultural Research Project
(K. Venkateswarlu, K. Vidyasagar Rao and B.S. Chandel)

The project is taken up to develop a statistical model for measuring the impact of NARP in 120 agro-climatic zones. The data collection is in progress.

11. Estimation of the Relation between Technology Development and Technology Adoption in Indian Agricultural Research System
(VE. Sabarathnam)

The study was undertaken to estimate the relation between technology generated and technology adopted in different agro-climatic zones in the country, and to find out the relationship between technologies generated by research institutes and the problems of the farming community.

During the year under report, the relation between the technology developed and adoption for each institute was worked out. Farmers were selected and asked to indicate the technologies adopted by them in the period year. In certain villages reasons for adoption of a single technology or multiple technologies by the farmers, among the technologies adopted, were also identified. The overall relationship between technology development and technology adoption for the Indian Agricultural Research System is being worked out.

12. Rationalization of Indigenous Technologies in Indian Agriculture (VE. Sabarathnam)

The study was undertaken to collect information on the various indigenous technologies that are being practised by the farmers, for the benefit of extension workers and the researchers engaged in agriculture.

During the year under report, the indigenous technologies from states like Tamil Nadu, Himachal Pradesh, West Bengal, Uttar Pradesh and Orissa were collected. Their relationship and utility value have also been assessed.

3.3 Education

Summer Institute on Recent Advances in Agricultural Research Management

In order to sensitize the senior research managers of SAUs and ICAR Institutes on recent developments in concepts and techniques of various aspects of Agricultural Research Management, a summer institute on Recent Advances in Agricultural Research Management was organized at the Academy from June 3 to 22, 1996.

The course covered various areas like ARM concepts, process, and developments; agricultural research project management - principles, techniques, and applications; ARM information system; manpower planning for effective agricultural research system; technological forecasting and systems approach, etc. A blend of lectures, group discussions, exercises and case analysis were used to cover different aspects during the course.

Dr S.N. Saha, Principal Scientist, was the Director of this Summer Institute. Twenty seven scientists belonging to the ICAR Institutes and SAUs participated in the programme.

Outreach Programme in Educational Technology

The Academy offers one-year post-graduate diploma programme in Educational Technology (OPET). The OPET programme even though offered based on the philosophy of distance education, is different from the other conventional distance education programmes where only print medium is used as carrier of instruction. This programme facilitates incorporation of essentials of educational technology in the Agricultural University education. The principle objective of the OPET is to offer a orientation programme to all faculty members of SAUs and ICAR deemed to be Universities covering all aspects of higher agricultural education, more specifically in educational methodology and instructional technology, to produce appropriately trained manpower who could serve the needs of these institutions by way of organizing in-service training programmes for teachers, undertaking education research, contributing to the upgradation of educational programmes, learning-teaching activities, etc. and by encouraging the Agricultural University teachers to conduct research in learning-teaching, testing and evaluation, and other allied fields of agricultural education.

After the successive completion of two such programmes, the third OPET programme was initiated with 22 participants. For each programme, there were three contact sessions - two in the I phase (one week of orientation in the beginning and another of two weeks in middle) and one in the II phase (two weeks at the end). Three contact sessions are compulsory, and are meant to train in instructional skills, preparation of resource materials like audio tutorials, computer assisted programmed lessons, self-instructional modules, evaluation materials, etc.

During the year under report, the third contact session of the third outreach programme was conducted from June 18 to 28, 1996. Eighteen academicians of various cadres from SAUs attended the programme. The programme is being coordinated by Dr A. Gopalam and Dr K.M. Reddy.

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3.4 Workshops

To facilitate the exchange of information and interaction among the scientists of the NARS, the Academy conducts a number of workshops. These workshops address the current and emerging areas of interest in the field of agricultural research management. During the year under report, two workshops were conducted, the details of which are given below:

National Workshop on PR Oriented Marketing Communication Strategies of Agricultural Research

Every organization should devise a sound marketing communication strategy, which includes aspects like public relations, marketing, advertising, sales promotion, personal selling, etc. Keeping in view the importance of all these factors, a national workshop on PR Oriented Marketing Communication Strategies of Agricultural Research was conceived and organized at the Academy from September 2 to 7, 1996.

The workshop was organized with the aim of exposing the participants to the importance of PR oriented marketing communication strategies in agricultural research. During the workshop issues like marketing communication, integrated brand communication, corporate PR, marketing strategy, customer orientation, transparency and visibility, media relations, multimedia in brand communication, etc. were discussed.

PR professionals belonging to the ICAR Institutes, SAUs, and banking sector, and business executives and entrepreneurs from private sector participated in the workshop. The co-ordinators for this workshop were Mr Abhijit Bhattacharjee and Dr Jagdeesh C. Kalla. Prof. Ch. Shyamsunder from Osmania University, Hyderabad acted as a facilitator in this workshop, which was attended by 24 delegates.

Training-cum-Workshop on Participatory Methods for Identification and Diagnosis of Research and Development Problems

The success of any development plan in agriculture and rural development depends not only on the support of research and extension services but also on peoples' participation. The basic philosophy of the Participatory Technology Development (PTD) and Participatory Rural Appraisal (PRA) is to empower rural people to improve

their standard of living. The PRA and PTD techniques aim to document and analyse diverse rural settings and processes that aid in planning; and for the past decade and a half constant search is on developing relevant, location-specific PRA methods.

Realizing the need to develop necessary skills to choose from a cafeteria of PRA / PTD tools and techniques for location-specific problem solving, a training-cum-workshop on participatory methods for identification and diagnosis of research and development problems was organized from February 3 - 7, 1997 at the Academy. The topics covered were -PRA/PTD - basic philosophy and genesis; PRA tools and techniques like direct observation techniques, analytical tools and techniques, individual and focus group interactions, participatory mapping/ranking, etc. ; PRA for watershed development; PRA for women concerns in agriculture; PRA for rural entrepreneurial development; case studies of PRA; analysis and interpretation of PRA observations and data; development of location specific action plans based on PRA-PTD techniques.

Eleven senior level faculty members and scientists from SAUs and ICAR Institutes, *Krish! Vigyan Kendras*, district and block-level extension personnel and development executives from State Department of Agriculture, and development executives from public and private sector, and non-governmental organizations attended the course. Dr N. Sandhya Shenoy and Dr S.N. Saha were the co-ordinators.

3.5 Consultancy

Faculty Development Programme in Educational Technology

Considering the expertise developed at this Academy in the area of educational technology and management, this Academy has undertaken the major HRD activity through faculty development programme to all the 28 State Agricultural Universities in the country, and four deemed-to-be Universities of ICAR. The new package of faculty development programme has been developed for SAUs as per their needs. The following faculty development programmes were organized by the Academy during the period under report:

S.No.	Programme	Period	Place
1.	Faculty Development Programme in Educational Technology	August 5-20, 1996	Rajasthan Agric. University, Udaipur
2.	Faculty Development Programme in Educational Technology	October 9-24, 1996	Gujarata Agric. University, Anand
3.	Orientation and Refresher courses for TANUVAS	January 16-31, 1997	TANUVAS, Chennai
4.	Faculty Development Programme in Educational Management and Administration	March 5-14, 1997	TANUVAS, Chennai

The off-campus programmes were organized at the request of respective Vice chancellors of the concerned Universities. The programmes emphasized on the concepts of management of agricultural education, behavioural modification in learning and teaching, testing, evaluation, and curriculum development.

Consultancy to Fountain Renewable Resources Ltd., UK

Consultancy was provided to Fountain Renewable Resources Limited, UK, by Dr Jagdeesh C. Kalla, for the project on Rehabilitation of Common Lands in the Aravalli Hills, Haryana from April to June, 1996.

Consultancy to NRCS, Indore

Consultancy was provided to National Research Center for Soyabean (NRCS), Indore for production of Video on NRCS. Mr K.R. Prabhakar and his team extended the service with 15 days of shooting in the month of September 1996.

Consultancy to Other Training Institutes

The Academy provided expertise in the form of guest lectures to other local training and educational institutions such as the Extension Educational Institute, Home Science College of Acharya N.G. Ranga Agricultural University, SVP National Police Academy, Institute of Cooperative Management, Regional Station for Forage Production and Demonstration, WALAMTARI, NIRD and ICRISAT. Most of the faculty members of the Academy were invited to deliver guest lectures in the above mentioned institutions. Subjects covered in these lectures were from different research areas of management and educational technology.

3.6 Miscellania

Hindi Cell Activities

Two Hindi workshops were organized by the Hindi Cell of the Academy from December 17 to 20, 1996. The first workshop was organized for the employees of the Academy with the support of the faculty members of the Central Hindi Training Sub-Institute, Hindi Teaching Scheme, Hyderabad to make the employees competent in discharging their duties through Hindi. A total of sixteen employees were trained in this workshop.

Another workshop was organized for the Assistant Directors(Hindi), Hindi Officers, Technical Officers(Hindi), Translators, and Hindi Cell In-charges of the ICAR Institutes and other central government organizations and undertakings. The inaugural function was held on December 17, 1996, under the chairmanship of Dr P. Kumar In-charge Director. During his inaugural speech he said that the father of the nation, Mahatma Gandhi, realized the importance of Hindi and helped in establishing the Dakshina Bharatiya Hindi Prachara Sabha at Madras. People came forward voluntarily to learn Hindi in those days, he added. Mr Anil Kumar Dubey, Director (Hindi), ICAR, New Delhi, stressed the importance of the follow-up action on the decisions of the Hindi Committees. Dr A. Gopalam, Senior Scientist and Officer-in-charge, Hindi, gave a detailed workshop report. Dr A. Gopalam and Mr D. Venkateswarlu were the co-ordinators of the workshop.

Hindi week celebrations were organized at the Academy from September 12 to 19, 1996. The valedictory function was held on September 19, 1996, under the auspicious presence of a great freedom fighter and former Rajya Sabha Member, Mr Vandemataram Ramachandra Rao. While delivering the valedictory address, he stressed on the national integrity through the national language. He appreciated the increasing number of Hindi speaking people present in the southern states. He added that the opposition to Hindi is not from the people but from the politicians. While delivering the presidential address, Dr Jagdeesh C. Kalla, Director, said that the language of any nation is the soul of that country. Whether it is a philosophy or spiritual understanding, or a study of linguistic or ancient culture, language is necessary for understanding as well as for study, he asserted. Dr A. Gopalam, Officer in-charge, Hindi while presenting the report on the achievements of the Hindi Unit of the Academy during the year 1995 pointed out that the Academy won Rajbhasha Shield for the year 1995, which is being won by the Academy for the past four consecutive years from 1992 to 1995. Several competitions

like noting, drafting, elocution, essay writing, singing, etc. in Hindi were organized. Mr Vandemataram Ramachandra Rao, the Chief Guest, distributed the prizes to the winners of various competitions.

Guest Lectures

Dr K.V. Raman, distinguished professor from Cornell University, USA delivered a lecture on Biotechnology in Agriculture on September 17, 1996.

Mr T.J. Master, AVSM, Commandant, College of Defence Management, delivered a guest lecture on Decision Making to the trainee-scientists of the Academy on September 24, 1996.

Dr Challaiah, Director, Nagarjuna Agricultural Research and Development Institute, delivered a lecture on Role of Corporate Sector in Agriculture R&D on September 25, 1996.

NAARM - The South Zone Champion in Table Tennis

The NAARM contingents of sports, which participated in the ICAR inter-Institutional Zone IV Sports meet organized by Central Research Institute for Dryland Agriculture (CRIDA), Hyderabad from December 2 to 6, 1996 at RRC ground, Hyderabad, bagged the South zone Championship in men's table tennis event. The match was won by Mr Sham Bahadur, Mr S. Swamy, Mr Pandu Reddy and Mr V. Murali.

Dr (Ms) Sandhya Shenoy got the third prize in 100 m race. Dr (Ms) Kalpana Sastry got the third prize in 2 km cycling race. Dr (Ms) Sandya Shenoy, Dr Nikitha, Ms A. Vijayalakshmi and Ms Sessa Sai got the second prize in 4 x 100 M relay (women) race. Dr A. Debnath got the third prize in men's long jump event. The 39 member contingent, including seven ladies, was led by Dr A. Debnath.

Felicitations

Hon'ble Union Minister of State for Agriculture, Dr Ummareddy Venkateswarlu

Dr Ummareddy Venkateswarlu, Union Minister of State for Agriculture was felicitated by the ICAR scientists of Hyderabad on June 29, 1996 at the Academy. While addressing the scientists, Dr Venkateswarlu pointed out the deleterious effect of the imbalanced use of fertilizers on soil fertility. The problem has reached catastrophic proportions and if the trend continued most of the soils may go out of cultivation, he said.

Dr Venkateswarlu said that restructuring the fertilizer prices is the only answer. He expressed the hope that the normal soil health could be restored in four or five years, by regulating the use of fertilizers.

The Minister said that the national agricultural policy would be announced soon. He indicated the formation of commodity-wise boards like Mango Board in Andhra Pradesh, as the state accounted for maximum mango production. Similarly a Floriculture Board could be based in Karnataka, a Pineapple Board in north-east, and an Apple Board in Himachal Pradesh. He also expressed satisfaction over the returns on investments to agricultural research, and said that the country was able to export technology to 46 countries. The vast talent of the research scientists could be put together to greater use, he added.

Hon'ble Union Minister for Agriculture, Mr Chaturanan Mishra

"Science has no boundaries; unfortunately, some scientists put barriers to science under their own concept of profiteering, giving birth to intellectual property rights being advocated in developed countries", said the Hon'ble Union Minister for Agriculture Mr Chaturanan Mishra while addressing the scientists of the various ARS Scientists' Forum Units located in Hyderabad. The programme was organized at NAARM on February 23, 1997.

The Hon'ble Minister stressed the need for consolidation and re-orientation of technology generation, technology assessment, refinement, and transfer to have more visibility. He also said that the farmers should get the best technology at their doorsteps;

institute-village linkage programmes should be organized to upgrade the research, leading to development of graded technologies suiting the farmers belonging to a range of economic strata. He emphasized the need for austerity measures and the importance of judicious spending of the budget.

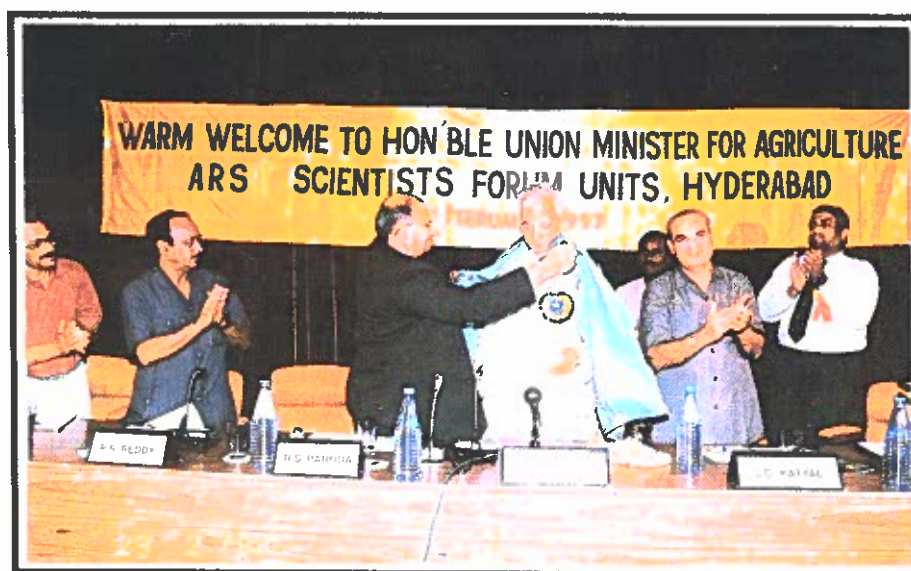
Dr R.S. Paroda, Director General, ICAR in his address, advised the scientists to improve their output leading to excellence in performance.

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Dr Jagdeesh C. Kalla, Director, NAARM, receiving the Raj Bhasha award from General Manager, South Central Railway.



Hon'ble Union Minister for Agriculture, Sri Chaturanan Mishra being felicitated by Dr R.S. Paroda, Director General, ICAR. Also seen Dr J.C. Katyal, Director, NAARM and Dr P.S. Reddy, Project Director, Directorate of Oilseeds Research.

4.1 Facilities

The Academy has good infrastructure facilities for carrying out the teaching - learning process more effectively and to help the trainees learn effective use of mass media for dissemination of information. The various facilities of the academic and miscellaneous nature available to the constituents of the Academy - trainees, faculty and staff are as follows:

4.1.1 Video Instruction and Production Laboratory

This Laboratory, comprising of two television studios and a master control room, facilitates intensive training of scientists in the preparation of effective resource materials for telecasting. It helps in conducting role play exercises for personality development. Video educational programmes are also produced. It is also useful for developing proper speech communication skills. Video shows were organized for various categories of visitors which include dignitaries from Govt. and Non-Govt. Organizations.

4.1.2 Audio-Visual and Photography Laboratory

This laboratory is equipped with all modern photographic and audio visual facilities, and it provides an excellent opportunity to gain experience in the preparation and use of audio-visual aids for effective presentation. These include overhead projectors, epidiascope, coloured transparency - making units, slide projectors, tape recorders and public address systems. Built-in sound Slide Projector with slide synchronizing facility. Audio cassettes duplicator, slide dissolve unit system and laminator machine. Audio-visual coverage was provided to seminars, workshops and classroom lectures organized at the Academy.

4.1.3 Speech and Mass Communication Laboratory

This Laboratory facilitates the scientists to record their programmes such as interviews, discussions and talks, specially for the rural listeners.

4.1.4 Organizational Behaviour Laboratory

Individual and group exercises in team-building, leadership and group dynamics conducted in this laboratory provide the trainees an opportunity to understand the various aspects of organizational behaviour including human thought, action and behaviour, and personality development.

4.1.5 Self-aided Instructional Laboratory

This laboratory provides opportunity for the participants to undergo structured and programmed instruction under self-learning situations. Automated electronic equipments and computerized instruction facilitate the process. A set of carrels with Telex Caramate imparting a tape cum slide programme, a VCP with monitor for video tutorial, and a computer for a programmed instruction are used. Software developed for learning Plant Pigments/Chlorophylls is being evaluated for its effectiveness as a learning tool.

4.1.6 Information Centre

The Academy has recognized its Information and Documentation Unit as Information Centre (IC). The vision of the IC is to be a provider of information through services and products that are delivered to the user, introduce and provide state-of-the-art information technology and be proactive in all its tasks.

In order to meet this vision, the IC is now structured into Book, Journal, Digital Media, and Field Experience Training (FET) and Foreign Tour (FT) reports, Sections, and Micrography and Reprography Units.

The Book section has collection of about 20,000 documents related to agricultural research, education, extension, and management. General access is provided to all documents. The entire collection is available through a computer-based catalogue through CDS/ISIS. The section also provides access to several computer based information systems such as Handistatus, Women in Agriculture in India, AGRUNIS, Rural Technology, etc.

The Journal Section Subscribes to nearly 300 international and national journals. It receives newsletters, reports and other material from NARS institutions and several international agencies such as CGIAR, FAO, World Bank, IDS, ODI, UNESCO, IFPRI etc.

The Digital Media (DM) Section focuses on providing access to digital media such as on Diskette, CD-ROM and Internet based information tools. At present the IC subscribes to Current Contents on Diskette (Agriculture and Social Sciences), AGRIS, AGRICOLA, AGRISEARCH, EMERALD ANBAR, TROPAG & RURAL CD-ROM Databases. The DM section also has several multimedia presentations. The DM Section has a dedicated GISTNIC terminal through which National Informatics Centre (NIC) databases can be accessed.

The FET and FT reports section is a repository of all FET reports of FOCARS and FT reports of agricultural scientists deputed on tours abroad.

The IC has full micrography production and access facilities and equipment, as also high speed, large volume reprography facility for duplication.

The IC has also initiated research in informatics. The Learning Resources Laboratory is now being developed to conduct research on techniques to convert information to knowledge in the context of agricultural research and related subject. It is being equipped with modern information processing, presentation tools, etc. The Information Management Laboratory's focus is on research in the use of new information technologies such as CD-ROM, DVD, Internet and Internet tools in accessing and disseminating information related to agricultural research, education, extension, and office administration and management.

In order to be user-centered in its services, the IC provides:

- Short and long-range referral services for data and information related to agricultural and management
- Documentation services through bibliographic searches
- Current awareness and selected dissemination of information based on user defined profiles
- Micrography access
- Reprography services
- GISTNIC access

The IC, as a part of its activities, also produces:

- Value added bibliographies
- Agriculture Research Management Abstracts - ARMA on Diskette
- Newspaper and management clippings

- Union catalogue of journals available in various libraries
- Suggested reading lists

In addition to providing information services and products, the IC also plays a major role in NAARM's education, training and skills enhancement programmes. These include:

- FOCARS
- Specialized Courses on Management of Information Centres and Information Management
- User education programmes
- Specialized courses for librarians and information scientists
- Outreach programmes
- Information Centres related consultancy

4.1.7 Computer Laboratory

The Computer Laboratory (CL) of the Academy has been reorganized after having been provided additional computing facilities under the ARIS Programme. The CL and the Local Area Network (LAN) is now being structured to be user-centered, providing state of the art LAN connectivity and stand-alone computing to NAARM faculty, staff and participants in various training programmes.

Under ARIS, the CL has received an HCL Meteor LAN Server and a Sun Sparc Computer. The LAN provides two operational platforms of Novell 4.1 Netware and Unix to all nodes that operate on Windows 95. Microsoft Office is the standard office suite used for documentation by all users in NAARM. In addition there is a variety of special purpose software for relational database management, statistical analysis and office automation.

The CL is also a training facility to provide training in computer use, managing information and information systems as also for participants of training programmes where there are areas of computer applications. The participants are from NARS Institutions.

There is an on-going activity to provide an Internet-based information system for NAARM, and Internet connectivity and E-Mail accounts to individual faculty.

4.2 Services

4.2.1 Administration, Finance and Accounts

This wing of the Academy is organized into five sections : Administration I deals with establishment matters; Administration II with cash and bills; Administration III with stores and purchases; Administration IV with estate management and training; and all matters pertaining to expenditure and bank reconciliation statements, monthly and annual accounts, budget and revised estimates, and internal audit are dealt with by the audit and accounts section.

This wing provides administrative, financial, and accounts support to all the Units of the Academy. Further, the Chief Administrative Officer and the Finance and Accounts Officer function as regular faculty members in delivering lectures on various rules and regulations of ICAR to ARS scientists and other trainees.

4.2.2 Farm Services

The Academy has developed the Agro-technology Demonstration Blocks (ATDB) in an area of 30 hectares, which serves as the show window of the NARS. The concept of ATDB is a unique innovation where the technologies developed and released by different ICAR Institutes and SAUs are demonstrated at one place for the benefit of participants of various national and international programmes.

4.2.3 Production and Publication Services

The Production and Publication Cell undertakes publication of various materials developed by the faculty for training purposes. The cell is engaged in production of books, annual report, folders, hand-outs, leaflets, information bulletins of the Academy and other works such as registers, stationery forms, letter heads, cover pages, invitation cards, etc. RICOH 6950 automatic plain paper copier was also acquired to enhance the production activities at faster pace. The cell is also equipped with Desk Top Publishing (DTP) System with several packages including MS Word, MacWrite, MacPaint, MacDraw, Page-Maker, and Hindi Font. This cell is equipped with an Offset Printing Press (Swift 300 Delux), processing camera, plate making, screen printing, and binding systems. The important publications brought out are NAARM News, Annual Report, NAARM Deepika, recommendations of various Workshops and Agricultural Research Management Abstracts (ARMA).

4.2.4 Hostel Services

The Academy has good board and lodging facilities for its visitors and trainees. The Academy's programmes are fully residential, and the training participants are accommodated in the Halls of Residence and in the Scientists' Home. These residential facilities have excellent catering services, which provide Indian, Chinese, Mughlai, and Continental cuisine. The Halls of Residence contains 154 fully furnished single occupancy rooms with attached bath. The Scientists' Home has 32 suites. Each suite is furnished with two beds, sitting and study areas, and an attached bath. Most of these suites are air-conditioned. There is a public call office in the Halls of Residence with STD/ISD facility. Colour television was installed in the dining hall for recreation of the trainee scientists. Believing that "health is wealth", gym equipment was also purchased for the physical exercise purposes for both trainees and staff of the Academy.

4.2.5 Banking Service

The Central Bank of India, Golconda Fort Branch, Hyderabad has been operating an Extension Counter (regular) in the faculty- cum-laboratory building to cater to the banking requirements of faculty members, staff and trainees. The counter also offers safe deposit locker facility to the faculty and the staff members.

4.2.6 Health Services

The Academy has a Health Center with a Medical Officer and a compounder. The center provides out-patients care to all probationers, faculty, staff and their families during scheduled hours. Emergency care is provided round-the-clock. One ECG analyzer, blood sugar estimator, oxygen instrument and suction apparatus were provided to the center for better medical care and treatment of the patients.

4.2.7 Public Relations Cell

The Public Relations Cell continued to engage in activities like effective linkage with the media to focus attention on the activities and achievements of the Academy. The cell, besides building the image of the Academy, also provides faculty support in the areas of publicity, visitors' coordination, graphics art aid and arranging the Academy's guest lectures. The cell also arranges cultural programmes, field trips and sightseeing programmes.

4.3 Other facilities

4.3.1 Residential Facilities

The Academy has a total of 95 residential quarters of different categories for the purpose of accommodating scientists, technicians, and staff of NAARM and other ICAR institutions in Hyderabad.

4.3.2 Recreation and Sports Facilities

The recreation facilities provided include several televisions connected to a dish antenna, common rooms with indoor games, and courts for tennis, volley ball and badminton.

TECHNICAL INFORMATION

5.1 Finance and Accounts

The budget estimates (1996-97) and revised estimates (1996-97) and expenditure during 1996-97 are given in the following table. The table also gives in detail expenditure on various Heads of Accounts

NON-PLAN

Sl. No.	Head of account	Budget estimates 1996-97 Rs.	Revised estimates 1996-97 Rs.	Expenditure during 1996-97 Rs.
1. i)	Establishment charges			
	a. Salary of officers			
	b. Salary of establishment			
	c. Dearness allowance	1,12,00,000	1,18,00,000	1,15,56,214
	d. House rent allowance			
	e. City comp. allowance			
	f. Other allowance / hon.			
	g. Interim relief			
	h. Bonus			
	i. Leave salary & pension contribution			
	Total	1,12,00,000	1,18,00,000	1,15,56,214
	ii. Overtime allowance	21,000	21,000	20,345
2.	Travelling allowance	1,60,000	2,00,000	1,90,547
3.	Other charges (excluding major construction works)	32,19,000	74,79,000	74,78,809
	Grand total	1,46,00,000	1,95,00,000	1,92,45,915

PLAN

Sl. No.	Head of account	Budget estimates 1996-97 Rs.	Revised estimate 1996-97 Rs.	Expenditure during 1996-97 Rs.
1.i) Establishment charges				
a.	Salary of officers			
b.	Salary of establishment			
c.	Dearness allowance			
d.	House rent allowance	32,50,000	14,25,000	13,69,583
e.	City comp. allowance			
f.	Other allowance / hon.			
g.	Interim relief			
h.	Bonus			
i.	Leave salary & pension contribution			
	Total	32,50,000	14,25,000	13,69,583
ii. Overtime allowance				
2.	Travelling allowance	7,00,000	9,00,000	8,85,310
3.	Other charges (excluding major construction works)	80,00,000	95,70,000	95,00,220
4.	Buildings and other works	1,05,50,000	64,05,000	63,52,325
	Grand total	2,25,00,000	*1,83,00,000	*1,81,07,438
RE sanction was Rs 200 Lakhs. Rs 17.00 Lakhs was surrendered to ICAR vide Letter no.15-3/96-97/ A&A dated 27.03.1997				

5.2 Human Resources

The human resources of NAARM is presented in the following table. Fifty per cent of scientific positions have fallen vacant.

S.No.	Name of the post	Sanctioned strength	Posis filled as on 31-3-1997
1.	Scientific	43	21
2.	Technical	66	64
3.	Administrative	50	47
4.	Supporting	41	41

5.3 Publications

Research Papers / Review Papers / Popular Articles

- Chandel, B.S. 1996. Estimating risk bearing ability of lower farm size groups for crop loans in Himachal Pradesh. *Agricultural Bankers* 19(4): 34 - 48.
- Chattopadhyay, C. and Kalpana Sastry, R. 1996. Management of sesame stem root rot through soil solarisation - an ecofriendly strategy. *Indian J. Myco. Pl. Path.* 26(1) : 123.
- Kalpana Sastry, R. and Chattopadhyay, C. 1996. Effect of soil polarization on population of *Fusarium oxysporum* f. sp. *carthami* in endemic soils. *Indian Phytopathology* 149 (in press).
- Kalpana Sastry, R. 1996. Symptoms of wilt disease - clues for use in resistance breeding. In: Proc. Training Prog. *Breeding Approaches for improving productivity in Safflower* (Hegde. et al. Ed.) pp 25-31.
- Muralidhar, U. and Rama Rao, D. 1996. Management information system - AGRUNIS : experiences and issues in Indian context. *Journal of Information Science* 22(4) : 277-286.
- Rama Rao, D., Muralidhar, U. and Kalla, J.C. 1996. Profile of scientific staff in agricultural universities in India. *European Journal of Extension* 3 (2): 119-129.
- Van Rheenen, H. A., Bhatti S. and Vidyasagar Rao, K. 1996. The sustainable preservation of Bio-diversity in self-pollinating plant species : sample size and collection methodology; *Journal of Genetics and Breeding* 96 (9).

Books / Reports

- Prasad, C. and Balaguru, T. (Eds.) 1997. *Women in Agriculture : Developmental Issues*. NAARM, Rajendranagar (In press)
- Suresh Kumar, M., Brahmaji, P.P. and Narasimha Rao, M. 1996. *Compilation on Agricultural Scientific Service Rules of ICAR*. NAARM, Hyderabad.

Venkateswarlu, K. and Jagdeesh C. Kaila. 1996. *SWOT Analysis of NARP*. Delta Publishing House, Hyderabad.

Book Review

Balaguru T. 1996. Conflict management - managing interpersonal conflict by K. Harigopal, Oxford and IBH Publishing Co. Pvt. Ltd., New Delhi. *Indian Journal of Training and Development* XXVI (3) : 91-94.

Papers Presented in Seminars / Conferences / Workshops

Balaguru, T. Agricultural research management in India : Role of NAARM. Paper presented in Management Programme Technical Seminar at ISNAR, The Hague, the Netherlands on December 10, 1996.

Balaguru, T. Policy, organization, and management issues in setting up an MIS. Paper presented in the ISNAR-JIRCAS Project Planning Meeting on 'New Technologies for Agricultural Research' on December 26, 1996 at JIRCAS, Tsukuba, Japan.

Jagannadham Challa, HRD Strategies for veterinary profession - an outlook, Invited paper for the Seminar on Veterinary and Animal Sciences Education and Research - Problems and Challenges at the Golden Jubilee Celebrations of the College of Veterinary Sciences, CCS HU, Hisar held on February 15 - 16, 1996.

Jagannadham Challa and Saha, S.N. Organizational behaviour and culture - a case for decision making. Case prepared for analysis and training for the International Management Development Programme on Management of Agricultural Research, held at NAARM, from February 3- 14, 1997

Jagannadham Challa and Saha, S.N. Agricultural research and education management - an emerging frontier for cooperation in South Asia. Paper presented in the International Seminar on Economic Cooperation among SAARC countries: Challenges for the Second Decade held at the Centre for SAARC studies, Andhra University, Visakhapatnam, from March 24 - 26, 1997.

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Nishi Sethi and Jagannadham Challa. Effectiveness of field practicals in teaching learning process. Paper presented in the International Conference on Information Technology in Education and Training, organized under the aegis of All India Association for Educational Technology at Indore, on December 28-30, 1996.

Sabarathnam, VE. Role of Extension, farmers and NGO's at each step of the A to Z steps of technology. Participated and presented in a National Workshop on Technology Assessment conducted by International Extension Forum at UAS, Bangalore from May 25 to 26, 1996.

Saha, S.N. Agricultural Education - Present scenario and impending problems. Participated and presented a paper at the meeting of the High Power Committee on Agricultural Education appointed by Govt. of India held at Dr M. S. Swaminathan Research Foundation, Chennai, on February 11, 1997.

Saha, S.N. Manpower planning by educational initiative in agriculture. Participated and presented a paper at the second meeting of the High Power Committee on Agricultural Education held at Punjab Agricultural University, Ludhiana from March 12 to 13, 1997.

Shanmugam, S. Study of perceived organizational climate, role efficacy and organizational role stress among scientists in the conference. Participated and presented a paper at the 2nd Asian and 33rd Indian Academy of Applied Psychology International Conference on Developing Human Relations and Ethnic Understanding conducted by Department of Psychology, Gurukul Kangri University, Haridwar from February 26 - 28, 1997.

Venkateswarlu, K. A case study in innovative location specific research in India. Presented a paper in the 4th AMDISA Academic Conference held at Sri Lanka from April 24 to 26, 1996.

Foreign Visits

- ** Dr K. Venkateswarlu, Principal Scientist & Unit-In-Charge, Project Management Unit attended the 4th AMDISA Academic Conference held at Sri Lanka from April 24 to 26, 1996.
- ** Mr M. Suresh Kumar, Chief Administrative Officer participated in the 1996 International Conference and Exposition organized by American Society for Training and Development held at Orange County Convention Center, Orlando, Florida, USA from June 1 to 7, 1996.
- ** Dr K. M. Reddy, Senior Scientist attended an international conference on Educational Technology 2000, organized by the Common Wealth of Learning, Canada and Asian Media Information and Communication Center, Singapore, held in Singapore from August 14 to 17, 1996.
- ** Dr T. Balaguru, Senior Scientist, and Mr U. Muralidhar, Systems Analyst visited ISNAR Netherlands from December 3 to 13, 1996, and participated in the four-day workshop on INFORM-R and a two-day training in Microsoft Access for Windows 95.
- ** Dr K. Vidyasagar Rao, Senior Scientist visited Israel and attended a course on Computers in Agricultural Production and Management from March 12 to April 9, 1997.
- ** Dr T. Balaguru, Senior Scientist, visited Japan and participated in the ISNAR-JIRCAS project planning Meeting on New Technologies for Agricultural Research held at Tsukuba, from March 24 to 27, 1997.

Advanced Training

Dr S. Shanmugam attended the 18th Faculty Development Programme in Management (FDPM) conducted by the Indian Institute of Management (IIM), Ahmedabad from June 27, 1996 to April 16, 1997. The courses taken by him include: Statistics for Management, Business Policy, Introductory Course on Information Systems, Management of Creativity and Innovation, Communications for Management Teachers, Foundations for Management, Research Methodology, Human Resources Management, and Organizational Behaviour. A project work was also undertaken on the Heuristics of personality, organizational climate, role efficacy and stress among scientists - A case study with special reference to ICAR.

5.4 Visitors

1. Ms Vibha Pandey, Director (Finance) visited the Academy on 18-04-1996. She inaugurated the second batch of ARFIS programme for the third batch at NAARM.
2. Mr N. Parthasarathy, Financial Adviser of ICAR visited the Academy on 21-04-1996 and inaugurated the ARFIS programme at NAARM on 21-04-1996.
3. Dr R.S. Paroda, Director-General of the Indian Council of Agricultural Research visited the Academy on 25-04-1996 and addressed the trainees.
4. A group constituting of 10 Deputy Superintendents of Police undergoing Induction Training Programme at the Andhra Pradesh Police Academy visited the Academy on 10-05-1996.
5. A group constituting of 11 participants (Senior Police Officials) undergoing 24th Training of Trainers Course at the National Police Academy visited the Academy on 16-05-1996.
6. Mr Michaela Wright, Director General XII European Commission and Mr. Marc De Bruycker, European Commission official from Brussels along with Mr. Sadhana Relia from DST, New Delhi visited the Academy on 28-05-1996.
7. A group of 60 final B.V.Sc. (CD batch) students of Madras Veterinary College, Tamil Nadu Veterinary and Animal Sciences University, accompanied by Dr.V.Purushothaman, Professor of Microbiology and Dr. Justin William, Assistant Professor of Surgery visited the Academy on 06-09-1996.
8. Mr Govert Gijsbers from ISNAR, The Netherlands visited on 10-9-1996.
9. A group of 63 students of III year B.Sc. (Ag.), College of Horticulture, Kerala Agricultural University, Thrissur, visited the Academy on 19-10-1996.
10. A group of 50 students of Final Year Agricultural Engineering from Annamalai University, Madras, accompanied by Dr Santha Govind, Assoc. Professor visited the Academy on 28-10-1996.
11. A batch of 61 students of B.Sc. (Ag.), Annamalai University, Madras led by Dr.K.Kanagasabapathi, Lecturer in Agricultural Extension visited the Academy on 06-11-1996

12. Dr M.A.Hamid Miah, Member Director (Crops) Bangladesh Agricultural Research Council visited the Academy on 13-11-1996.

13. Dr Derek R.Byerlee, Principal Economist, Agricultural and Natural Resources Department, World Bank visited the Academy on 11-11-1996.

14. Dr P.V. Shenoi, Director, Institute for Social and Economic Change, Bangalore and Consultant of the World Bank visited the Academy on 09-12-1996. He held detailed discussions with the Faculty Members with respect to organizational and managerial reforms in the ICAR system as a part of the forthcoming World Bank supported National Agricultural Technology Project.

15. A batch of 21 students of III B.Sc. (Hort.) of the Dr Y.S. Parmar University of Horticulture and Forestry from Nauni, Solan (H.P.) visited the Academy on 18-12-1996 as part of their educational tour.

16. A batch of 33 students of III B.Sc. (Ag. Engg) of the Dr Y.S. Parmar University of Horticulture and Forestry from Nauni, Solan (H.P.) visited the Academy on 24-12-1996 as part of their educational tour.

17. Mr J.B.S. Umanadh, Senior Business Reporter from NEWSTIME, a popular local English daily of Hyderabad visited the Academy on 09-01-1997

18. Hon'ble Union Minister of Agriculture, Sri Chaturanan Mishra visited the Academy and addressed the scientists of the ICAR institutes located in Hyderabad on 23-02-1997.

19. An international delegation of 40 ICRISAT Governing Board members visited the Academy on 24-02-1997.

5.5 Committees

Management Committee

The Academy is guided in planning, organizing and conducting its various programmes by the Management Committee. The Management Committee approves the programmes to be undertaken. The Academy ensures their implementation, both in terms of physical targets and schedules, and provide broad guidelines for future programmes. The Management Committee also helps in formulation and consideration of proposals for Five Year Plan and Annual Plans and takes up periodical review of progress of development schemes.

The Management Committee, besides assisting Director of the Academy in the discharge his functions; focuses its attention on the research and other programmes of the Academy, and ensures their effective implementation. While examining these programmes periodically, it pinpoints bottlenecks, if any, and suggests remedial measures.

The 27th and 28th meetings of the Management Committee of the Academy were held on September 26, 1996 and January 29, 1997. The composition of the Management Committee of NAARM is as follows :

- | | | |
|----|---|---|
| 1. | Dr Jagdeesh C. Kalla
Director, NAARM, Hyderabad. | Chairman (till Nov. 30, 1996) |
| | Dr S.N. Saha
Director-in-Charge | Chairman (from Dec. 1, 1996 to Feb. 19, 1997) |
| | Dr J.C. Katyal
Director | Chairman (from Feb. 20, 1997) |
| 2. | Dr M.V. Rao
Vice Chancellor, APAU, Hyderabad. | Member |
| 3. | Dr J.C. Katyal
Director, CRIDA, Hyderabad. | Member |
| 4. | Dr Harpal Singh
Dean, P.G. Studies
G.B. Pant University of Agriculture &
Technology, Pantnagar, U.P. | Member |

- | | | |
|-----|---|------------------|
| 5. | Mr V. Ananda Rao
Principal Secretary
Dept of Agriculture, Govt. of A.P. | Member |
| 6. | Mr P.V. Eapachan
Finance & Accounts Officer
Central Research Institute for
Dryland Agriculture, Hyderabad. | Member |
| 7. | Mr Mohd Haneef Khatri,
General Secretary
Dist Congress Committee
Rajasthan. | Member |
| 8. | Ms K. Bharathi Devi
Social worker, Kakinada. | Member |
| 9. | Dr O.P. Pareek
Director,
NRC for Arid Zone Horticulture
Bikaner, Rajasthan. | Member |
| 10. | Dr D. Jha
Director, NCAP
IASRI Campus
New Delhi. | Member |
| 11. | Dr Mangala Rai
Deputy Director General (P)
ICAR Krishi Bhavan
New Delhi. | Member |
| 12. | Mr M. Suresh Kumar
Chief Administrative Officer
NAARM, Hyderabad. | Member-Secretary |

Research Advisory Committee

The Research Advisory Committee (RAC) of NAARM was constituted by the ICAR with the sole purpose of suggesting relevant research programmes in the area of Agricultural Research and Educational Management. The RAC comprises the following senior level research managers and scientists

- | | | |
|----|---|-----------------------------|
| 1 | Dr Maharaj Singh
Ex-DDG (Education)
New Delhi | Chairman |
| 2. | Deputy Director General (Education)
ICAR, New Delhi | Member |
| 3. | Dr M.S. Chari
Director, CTRI
Rajahmundry | Member |
| 4. | Dr Harpal Singh
Dean, G.B. Pant University
Pant nagar | Member |
| 5. | Dr Y.L. Nene
DDG, ICRISAT
Hyderabad | Member |
| 6. | Dr Dharani P. Sinha
President and MD, COSMODE
Hyderabad | Member |
| 7. | Dr K.S. Ramesh
Dean, ASCI
Hyderabad | Member |
| 8. | Dr Anil K. Gupta
IIM, Ahmedabad | Member |
| 9. | Dr Jagdeesh C. Kalla
Director, NAARM | Member (Till Nov. 30, 1966) |

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| 10. | Dr S.N. Saha
Director-in-Charge | Member (From Dec.1, 1996 to Feb.19,1997) |
| 11 | Dr J.C. Katayal
Director, NAARM | Member (From Feb.20, 1997) |
| 12. | Dr T. Balaguru
Sr. Scientist, NAAR | Member-Secretary |

Staff Research Council

The Staff Research Council (SRC), deliberates on the research activities in the areas of agricultural research and education management. The Council consists of the following members.

- | | | |
|----|--|---|
| 1. | Dr Jagdeesh C. Kalla
Director | Chairman (till Nov.30, 1996) |
| | Dr S.N. Saha
Director-in-Charge | Chairman (from Dec. 1, 1996 to
Feb.19, 1997) |
| | Dr J.C. Katyal
Director | Chairman (from Feb. 20, 1997) |
| 2. | Dr P. Kumar
Principal Scientist | Member |
| 3. | Dr S.N. Saha
Joint Director | Member |
| 4. | Dr K. Venkateswarlu
Principal Scientist | Member |
| 5. | Dr A. Gopalam
Senior Scientist | Member |
| 6. | Dr K.M. Reddy
Senior Scientist | Member |

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7.	Dr M.M. Anwer Senior Scientist	Member
8.	Dr K. Vidyasagar Rao Senior Scientist	Member
9.	Dr P. Manikandan Senior Scientist	Member
10.	Dr Jagannadham Challa Senior Scientist	Member
11.	Dr S. Shanmugam Senior Scientist	Member
12.	Dr VE. Sabarathnam Senior Scientist	Member
13.	Dr D. Rama Rao Senior Scientist	Member
14.	Dr A.U. Gadewar Senior Scientist	Member (Till Aug. 7, 1996)
15.	Mr U. Muralidhar Systems Analyst	Member
16.	Dr T. Balaguru Senior Scientist	Member - Secretary

During 1996-97, four meetings were conducted to finalize the research programmes of the Academy. In the process, several projects pertaining the HRD, Project Management, TOT and other areas relevant to research management were initiated.

Academic Council

The Academic Council (AC) oversees all the academic functions of NAARM. The following was the composition of the AC during 1996-97. The Council held two meetings during this year.

- | | | |
|----|---|---|
| 1. | Dr Jagdeesh C. Kalla
Director, NAARM | Chairman (Till Nov. 30, 1996) |
| | Dr S.N. Saha
Director-in-Charge | Chairman (from Dec.1, 1996 to
Feb. 19, 1997) |
| | Dr J.C. Katyal
Director, NAARM | Chairman (from Feb. 20, 1997) |
| 2. | Dr P. Kumar
Principal Scientist | Member |
| 3. | Dr A.U. Gadewar
Senior Scientist | Member (Till Aug.7, 1996) |
| 4. | Dr A. Gopalam
Senior Scientist | Member |
| 5. | Dr M.M. Anwer
Senior Scientist | Member |
| 6. | Ms Suguna Sri
Librarian | Member (Till Sept. 10, 1996) |
| 7. | Dr S.N. Saha
Director-in-Charge | Member-Secretary |

Official Language Implementation Committee

The Official Language Implementation Committee of the Academy provides periodical suggestions for implementation of Official Language Policy, and rules and regulations of government of India that are issued from time to time. The Committee met four times during the year on June 28, September 30, December 31, 1996 and March 29, 1997, and discussed several issues relating to the implementation of Hindi language. The following were the members :

- | | | |
|----|----------------------------------|-------------------------------|
| 1. | Dr Jagdeesh C. Kalla
Director | Chairman (till Nov. 30, 1996) |
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	Dr S.N. Saha, Director-in-Charge	Chairman (from Dec.1,1996 to Feb.19,1997)
	Dr J.C. Katyal Director, NAARM	Chairman (from Feb. 20, 1997)
2.	Dr A. Gopalam Senior Scientist	Vice-Chairman
3.	Dr D. Rama Rao Senior Scientist	Member
4.	Mr M. Suresh Kumar Chief Administrative Officer	Member
5.	Mr B. Pandu Reddy Finance & Accounts Officer	Member
6.	Mr K.R. Prabhakar Programme Officer (CCTV)	Member
7.	Mr U. Muralidhar Systems Analyst	Member
8.	Mr R.V.V.S. Prakasa Rao Editor	Member
9.	Mr Abhijit Bhattacharjee Public Relations Officer	Member
10.	Ms Jhansi Laxmi, Sr. P.A.	Member
11.	Ms G. Aneeja, Asst. Editor	Member
12.	Mr D. Venkateswarlu Tech. Officer (Hindi)	Member-Secretary

5. 6 Personnel

1. Dr Jagdeesh C. Kalla, Director (till Nov. 30, 1996)
Dr S.N. Saha (JD), Director-In-Charge (from Dec.1,1996 to Feb.19, 1997)
Dr J.C. Katyal, Director (from Feb. 20, 1997)

Agricultural Research Systems and Policies Unit

2. Dr P. Kumar, Principal Scientist & Unit In-Charge
3. Dr VE. Sabarathnam, Senior Scientist
4. Dr T. Balaguru, Senior Scientist
5. Dr K. Vidyasagar Rao, Senior Scientist
6. Ms A. Vijaya Lakshmi, Stenographer
7. Ms A. Mercy, Stenographer
8. Mr M. Shyam Rao, SSG-1

Educational Systems and Technology Unit

9. Dr A. Gopalam, Senior Scientist & Unit In-Charge
10. Dr K.M. Reddy, Senior Scientist
11. Dr D. Rama Rao, Senior Scientist
12. Mr M. Venkatesh, Stenographer
13. Mr M. Yadalah, SSG-II

Transfer of Technology Systems and Policies Unit

14. Dr S.N. Saha, Joint Director (from Aug. 9, 1996)
15. Dr A.U. Gadewar, Senior Scientist (till Aug. 8, 1996)
16. Mr R.C Chelvan, Scientist (SG) (on study leave)
17. Mr N.R. Nageswara Rao, Tehnical Officer (T-5)
18. Mr L. Venkateswarlu, Photographer (T-4)
19. Mr Bansidhar Nayk, Projectionist (T-II-3)
20. Mr D. Rajagopal Rao, PAE Operator
21. Mr S. Sunder Raj, Media Operator (T-1)
22. Ms T. Vanisri, Stenographer
23. Mr G. V. Bikshapathi, SSG-II

Project Management Unit

24. Dr K. Venkateswarlu, Principal Scientist & Unit In-Charge
25. Dr P. Manikandan, Senior Scientist
26. Ms L. Jhansi Lakshmi, Senior Personal Assistant
27. Ms K.K. Rukmani Ammal, Junior Stenographer
28. Mr K. Satyanarayana, SSG-II

Human Resources Development Unit

29. Dr M.M. Anwer, Senior Scientist & Unit In-Charge
30. Dr Jagannadham Challa, Senior Scientist
31. Dr S. Shanmugam, Senior Scientist
32. Ms S. Sessa Sai, Junior Stenographer
33. Mr N. Narasinga Rao, SSG-I

Information and Documentation Unit

34. Ms M. Suguna Sri, Librarian & Unit In-Charge (till September 10, 1996)
35. Ms P.V. Nirmala, Technical Officer (T-5)
36. Mr Y.V. Siva Prasad, Documentation Assistant (T-4)
37. MS Y. Dhanalakshmi, Library Assistant (T-2)
38. Ms S. Shanti, Junior Stenographer
39. Mr Venkatesham, Xerox Operator
40. Mr M. Ashok, SSG-II

Director's Cell

41. Mr N. Raghunath, Stenographer
42. Mr T.V. Ram Das, Stenographer
43. Mr P. Swamy, SSG-II

Training Cell

44. Ms Sarada Samanta, Senior Stenographer
45. Mr P.Vijender Reddy, Research Assistant (T-4)
46. Mr K. Hemlal, SSG-I

Hindi Section

47. Mr D. Venkateswarlu, Technical Officer (T-5)

48. Mr S. Pradeep Singh, Asstt Director (Hindi Instruction)
49. Ms J. Renuka, Hindi Assistant
50. Ms Bokde R. Shamrao, Hindi Typist

Computer Laboratory

51. Mr U. Muralkhar, Programmer (Computer)
52. Mr M.J. Pradeep Kumar, Asstt Programmer
53. Mr K. Venkateswara Kumar, Statistical Assistant (T-4)
54. Mr P. Mohan Singh, Computer Assistant (T-II-3)

Video Instruction and Production Laboratory

55. Mr K.R. Prabhakar, Programme Officer (CCTV)
56. Mr Ch. Janardhan Rao, Technical Officer (T-5)
57. Mr N. Naresh Kumar, VCR-Tele Cine Operator (T-I-3)
58. Mr S.P. Ravi, Sound Recordist
59. Mr C. Bikshapathi, SSG-II

Production and Publication Cell

60. Mr R.V.V.S.Prakasa Rao, Editor (T-7)
61. Mr N. Narasimha Reddy, Technical Officer (T-5)
62. Mr K. Anil Kumar, Reprographic Assistant (T-4)
63. Ms G. Aneja, Asstt Editor (T-II-3)
64. Mr Shekar Reddy, Dark Room Assistant (T-I-3)
65. Mr P. Srinivas, Proof Reader
66. Mr S. Swamy, Assistant Gestetner Operator
67. Mr Prashant Gaikwad, Binder
68. Mr C. Chandramouli, SSG-I
69. Ms C. Narasubal, SSG-I

Farm Section

70. Mr V. Murali, Farm Superintendent (T-7)
71. Mr M.A. Basith, Technical Officer (T-5)
72. Mr K. Rangaswamy, Technical Officer (T-5)
73. Mr B. Veeraiah, Farm Assistant (T-I-3)
74. Mr Ahire Laxman Maharu, Hort. Tech. Assistant (T-II-3)
75. Mr K. Shivaiah, Technician Electrical (T-2)
76. Mr C.J. Samuel, Junior Clerk

77. Mr M. Padmaiah, Tractor Driver
78. Mr P. Iswari, Power Tiller Driver
79. Mr P. Balraj, SSG-II
80. Mr S. Jangaiah, SSG-II
81. Mr B. Subba Rao, SSG-II
82. Ms B. Bharatamma, SSG-II
83. Ms N. Sukunamma, SSG-I
84. Mr G. Pentaiah, SSG-I
85. Mr R. Sattaiah, SSG-I
86. Ms G. Anasuya, SSG-I

Hostel Services

87. Mr Zameer Ahmed, Manager (Hostel Services)
88. Mr Sham Bahadur, Catering In Charge
89. Ms Savithri Srinivasan, Mess Supervisor
90. Mr P. Bhaktavastalam, Junior Clerk
91. Mr K. Pentaiah, SSG-II
92. Mr P. Yadaiah, SSG-II
93. Mr Khalid, SSG-II
94. Ms B. Santhamma, SSG-II
95. Ms S. Sakunthala, SSG-II
96. Mr A. Yadaiah, SSG-I
97. Mr M. Krishnaiah, SSG-I
98. Mr C. Anjaiah, SSG-I
99. Ms K. Kalavathi, SSG-II
100. Ms M. Yadamma, SSG-I
101. Ms I. Bharathamma, SSG-I
102. Ms T. Jangamma, SSG-I
103. Ms Nazia Parveen, SSG-I

Administration

104. Mr M. Suresh Kumar, Chief Administrative Officer
105. Mr K. Janakiramaiah, Superintendent (Asstt Adm. Officer)
106. Mr Y. Sankar Rao, Superintendent
107. Mr P.P. Brahmaji, Superintendent
108. Mr P. Neelakantam, Assistant
109. Mr P.G. Kohad, Assistant
110. Mr M. Narasimha Rao, Assistant
111. Mr K. Prabhu Das, Assistant

112. Mr M. Dinesh, Assistant
113. Mr T. Srinivas, Senior Clerk
114. Mr G. Raj Reddy, Senior Clerk
115. Mr C. Phani Raj, Senior Clerk
116. Mr P. Anand Kumar, Stenographer
117. Ms B. Padma Saroja, Senior Clerk
118. Mr K. Srinivasa Rao, Senior Clerk
119. Mr P. Srinivasu, Junior Clerk
120. Mr M. Sridhar, Junior Clerk
121. Mr P. Venkatesh, Junior Clerk
122. Mr K.R. Ghanshyam, Junior Clerk
123. Mr M.K. Samson, Junior Clerk
124. Mr B.H. Dharma Raj, Junior Clerk
125. Ms N. Vijaya Lakshmi, Junior Stenographer
126. Ms V. Shailaja, Junior Stenographer
127. Ms P.D. Jessie Ecclicia, PMBX Operator-cum Receptionist
128. Mr P. Anantha Reddy, Technician (Instrument) (T-II-3)
129. Ms K. Swarajya Laxmi, Junior Library Assistant (T-2)
130. Mr J. Chandraiah, SSG-I

Public Relations Cell

131. Mr Abhijit Bhattacharjee, Public Relations Officer
132. Mr A. Murali Mohan Rao, Technical Officer (T-5)
133. Mr P. Namdev, Technical Officer (T-5)
134. Ms Y. Anuradha, Junior Stenographer

Civil Engineering Cell

135. Mr Sohail Ahmad Khan, Junior Engineer(Civil) (T-4)
136. Mr K. Veeranasiah, Carpenter
137. Mr N. Prabhakar Rao, Plumber

Transport Section

138. Mr K. Sattaih, Driver
139. Mr S.K. Peera, Driver
140. Mr M.M. Mohan Rao, Vehicle Mechanic (T-I-3)
141. Mr G. Muthyalu, Driver
142. Mr U.V. Ratnam, Driver
143. Mr N. Ashok, Driver

- 144. Mr T. Laxman, Driver
- 145. Mr B. Narayana, SSG III

Security Cell

- 146. Mr B.Ch. Satyanarayana, Estate-cum-Security Officer
- 147. Mr M. Srinivasa Rao, Pump Driver
- 148. Mr B.K. Venkat Ram, Pump Driver
- 149. Mr G. Dasaratha, SSG-I
- 150. Ms M. Eswaramma, SSG-I
- 151. Mr M. Phool Kumar, SSG-I
- 152. Mr L. Satyanarayana, SSG-I

Electrical Section

- 153. Mr L. Ramesh, Technical Assistant (Elec.) (T-I-3)
- 154. Mr K. Obulapathi, Technician Electrical (T-2)
- 155. Mr M.K. Sonkusare, Electrician (T-2)
- 156. Mr M. Mohan Rao, Technician Electrical (T-2)
- 157. Mr B.S.N. Murthy, Technician Electrical (T-2)
- 158. Mr M.K. Shamshuddin, Technician Electrical (T-2)
- 159. Mr A. Nageswara Rao, Lineman (T-2)
- 160. Mr B. Satyanarayana, Technician Electrical (T-1)
- 161. Mr D.R.S. Rao, Pump Driver

Audit and Accounts Section

- 162. Mr B. Pandu Reddy, Finance & Accounts Officer
- 163. Mr C. Bagaiah, Assistant
- 164. Mr S. Balakamesh, Junior Clerk
- 165. Mr R. Chandra Babu, Junior Clerk
- 166. Ms K. Radha Sujatha, Junior Stenographer
- 167. Ms Y. Gayathri, Junior Clerk
- 168. Ms G. Manibal, SSG-III

Health Centre

- 169. Dr (Major) A. Debnath, Medical Officer
- 170. Mr A. Venkateswarlu, Compounder

Research Managers Joined

J.C. Katyal - Director of NAARM

Dr J.C. Katyal, an eminent Scientist of Soil Sciences by profession, assumed charge as Director of the Academy on February 20, 1997

Dr Katyal has held many important positions in the ICAR and in international research centers. Prior to his joining the Academy, Dr Katyal was the Director of CRIDA, Hyderabad.

He was a consultant to FAO on fertilizer use and multiple cropping systems. He also acted as a consultant to World Bank for the project titled Indian Rainfed Farming Study. He was also an expert consultant to UGC for preparing the curriculum on Dry Farming for undergraduate courses.

Dr Katyal is the recipient of many prestigious awards. Some of the important ones include Dr Norman E. Borlaug award in 1995, for his contribution to soil science and dryland agricultural research, and Rafi Ahmed Kidwai award in 1996 for development of Farmer-friendly Seedling Implements. He is an expert member of various committees and panels. He has to his credit more than 180 research and review papers, besides being the author of two books.

S.N. Saha - Joint Director of NAARM

Dr S.N. Saha, assumed charge as Joint Director of the Academy on October 14, 1996. Earlier he was Principal Scientist and Head in the Transfer of Technology Systems and Policies unit and also in charge of Training Programmes of the Academy. He has been contributing to the training and research programmes of the Academy in a great measure, ever since he joined the Academy in 1984.

M.S. Chari - Officer-on-Special Duty

Dr M.S. Chari, joined the Academy as Officer-on-Special Duty on May 1, 1966 to look after the duties of the Joint Director. He worked for about five months and left the Academy on September 23, 1996 to join his parent organization, the Gujarat Agricultural university, Anand.

Staff News

The following personnel were promoted during the period under report.

Name of the person	Promoted	
	From	To
Mr K.Janaki Ramaiah	Superintendent	Asst Administrative Officer
Mr P.P.Brahmaji	Assistant	Superintendent
Mr K.Prabhudas	Senior Clerk	Assistant
Mr M.Dinesh	Senior Clerk	Assistant
Mr K. Srinivasa Rao	Junior Clerk	Senior Clerk
Ms B.Padma Saroja	Junior Clerk	Senior Clerk
Mr N.Raghunath	Stenographer	Senior Stenographer
Ms Y.Anuradha	Junior Stenographer	Stenographer

Deputation

1. Mr K.Srinivasa Rao, Senior Clerk, was relieved from duty on June 26, 1996 to join TOT projects (Zone-V) located at CRIDA, Hyderabad, as Superintendent (A&A).
2. Dr A.U. Gadewar, Senior Scientist, left the Academy on lien to take up the post of Professor at the National Institute of Bank Management, Pune on August 8, 1996.

Welcome

1. Mr S.P. Ravi joined as Sound Recordist on July 10, 1996.
2. Ms Bokde R.Shamrao joined as Junior Clerk (Hindi Typist) on July 7, 1996.
3. Mr V. Murali, Garden Superintendent, joined this Academy on request transfer basis on September 2, 1996 from IIHR, Bangalore.

Resignation

Ms M. Suguna Sri, Librarian of the Academy, was relieved from the service on September 10, 1996.

Farewell

A fond farewell was given to Dr R. K. Samanta, Senior Scientist of the Academy by the faculty members of NAARM, on his new assignment to the post of Zonal Co-ordinator, TOT Project, NDRI campus, Bangalore. He was relieved on March 6, 1997.

Retirement

Dr Jagdeesh C. Kalla superannuated on November 30, 1996, after a distinguished service of one and a half years as Director and five and a half years as Joint Director of this Academy.

During his tenure at the Academy, Dr Kalla has been actively associated with FAO-UNESCO-UNDP sponsored project on Advanced Centre for Agricultural Education and Research Management. Dr Kalla provided valuable guidance in developing course material, case studies, evaluation and monitoring studies. He participated and presented several papers and case studies in both national and international seminars and conferences. He is the recipient of the NCERT award given in recognition of his service to agricultural education.

